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Accelerating Mathematical and Scientific Discovery with Gemini Deep Think

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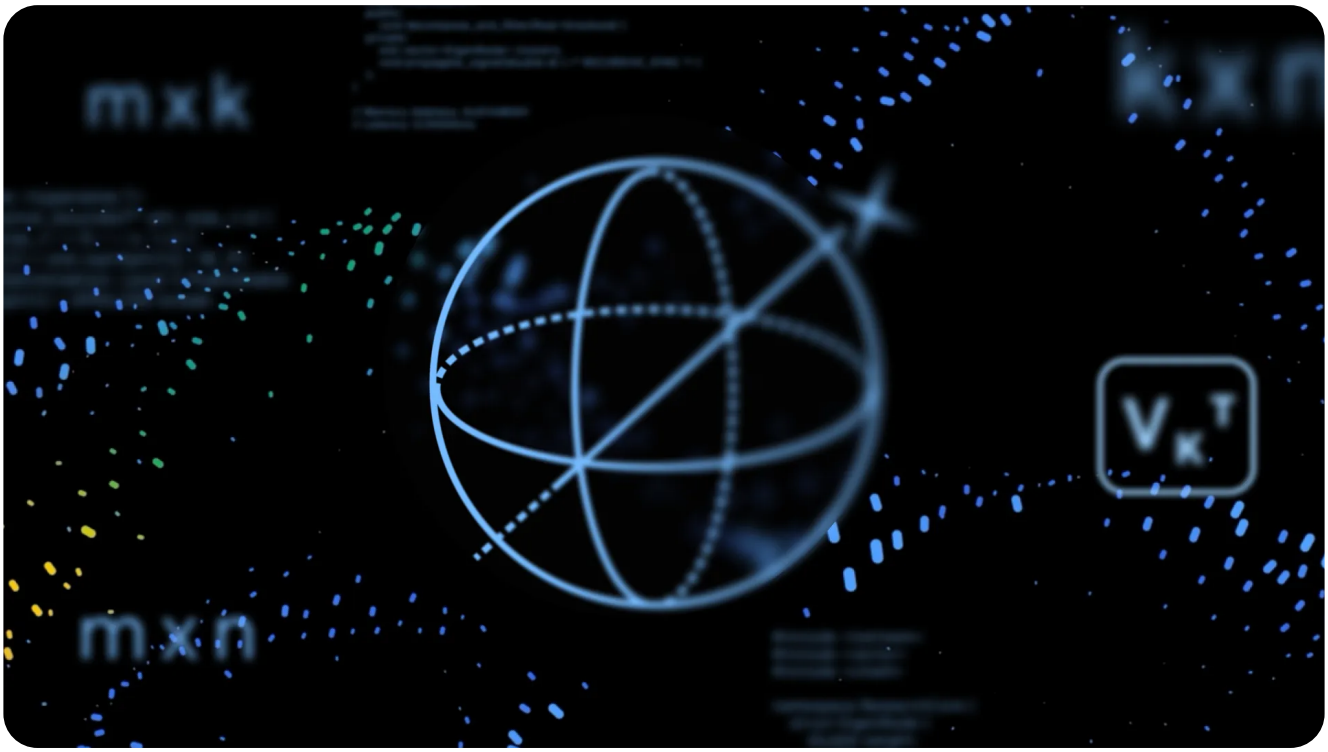

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Under direction from expert mathematicians and scientists, Gemini Deep Think is solving professional research problems across mathematics, physics, and computer science

In the summer of 2025, an advanced version of Gemini Deep Think achieved Gold-medal standard at the International Mathematics Olympiad (IMO) and later, an updated version, obtained similar results at the International Collegiate Programming Contest. These results demonstrated the model could reason through some of the most challenging math and programming problems designed for students. Since then, Gemini Deep Think mode has moved into science, engineering and enterprise workflows to tackle more complex, open-ended challenges.

In the last week, our teams published two papers

Deep Think mode. These results stem from deep collaboration between mathematicians, physicists, and computer scientists.

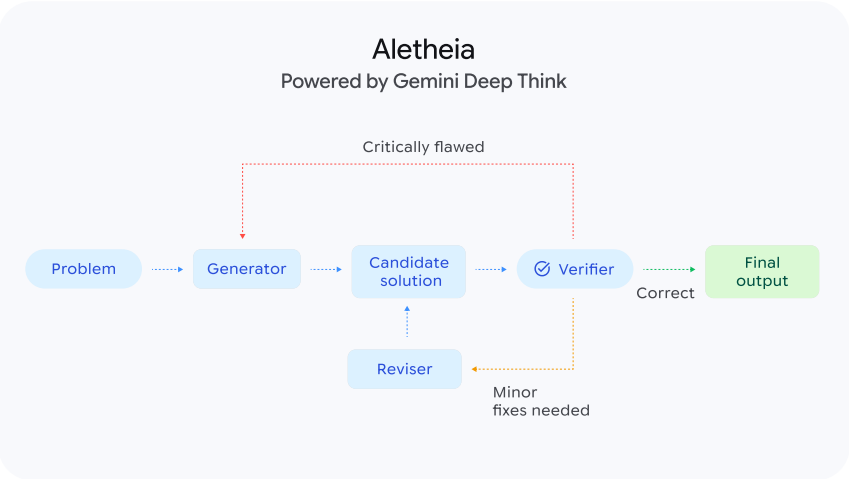
The Frontier of Pure Mathematics

Unlike IMO problems, research-level mathematics requires advanced techniques from vast literature. While foundation models have

large knowledge bases, data scarcity often leads to superficial understanding and hallucinations in advanced subjects.

To solve this, we built a math research agent (internally codenamed Aletheia), powered by Gemini Deep Think mode. It features a natural language verifier to identify flaws in candidate solutions and enable an iterative process of generating and revising solutions. Crucially, this agent can admit failure to solve a problem, a key feature that improved the efficiency for researchers.

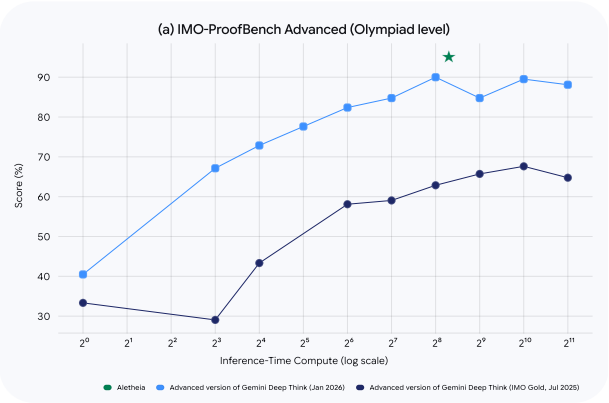
Additionally, the research agent uses Google Search and web browsing to navigate complex research, preventing spurious citations and computational inaccuracies when synthesizing published literature.



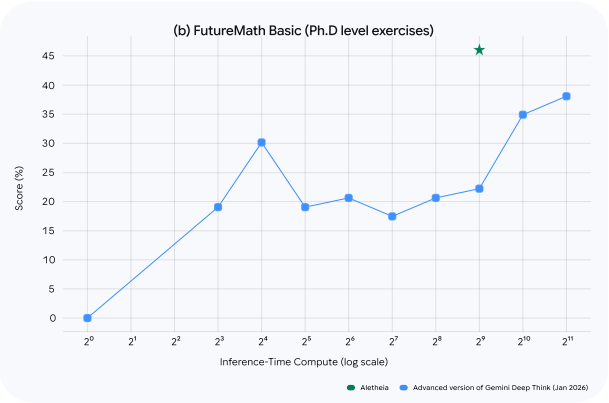
Overview of Aletheia, a math research agent powered by Deep Think that can iteratively generate, verify, and revise for research-level math problems.

Since achieving IMO Gold-medal standard in July 2025, Gemini Deep Think has progressed rapidly,

scoring up to 90% on the IMO-ProofBench Advanced test as inference-time compute scales. We demonstrated that the scaling law continues to hold as we progress beyond Olympiad level into PhD-level exercises (per our internal FutureMath Basic benchmark). Notably, Aletheia demonstrated that higher reasoning quality can be achieved at a lower inference-time compute.



The latest advanced version of Deep Think, as of Jan 2026, has significantly outperformed the IMO-Gold version (Jul 2025) on Olympiad-level problems. Aletheia makes further leaps in terms of reasoning quality with lower inference-time compute. All results were graded by human experts.



The inference-time scaling law also transfers to PhD-level exercises.

For research-level math, Aletheia has already enabled several advancements, produced via varying levels of autonomous research:

- Reliable autonomous research. A research paper (Feng26) generated by AI without any human intervention, which calculates certain structure constants in arithmetic geometry called eigenweights.
- AI-guided collaboration. A research paper

(LeeSeo26) demonstrating human-AI collaboration in proving bounds on systems of interacting particles called independent sets.

- An extensive semi-autonomous evaluation (Feng et al., 2026b) of 700 open problems on Bloom's Erdős Conjectures database, including autonomous solutions to four open questions listed there. On Erdős-1051, our model autonomously solved and helped lead to a generalization reported in a research paper (BKKKZ26).

The agent also contributed intermediate propositions on two further papers, (FYZ26) and (ACGKMP26). It is also of note that there has been prior work using Gemini for research-level math at a smaller scale in terms of collaborations and the number of problems tackled.

Following extensive discussions with the mathematical community, we suggest a taxonomy to classify AI-assisted mathematics research by significance and degree of AI contribution - contributing to the wider discussion on responsible documentation, evaluation and communication of AI-generated results. Level 2 ("publishable quality") works have been submitted to reputable journals. Currently, we do not claim any Level 3 ("Major Advance") and Level 4 ("Landmark Breakthrough") results.

	Human with secondary AI input	Human-AI Collaboration	Essentially Autonomous
Level 0: Negligible Novelty			Erdős-652, 654, 1040 (Feng et al., 2026b)
Level 1: Minor Novelty			Erdős-1051 (Feng et al., 2026b)
Level 2: Publishable Research*	Complexity Bounds (ACGKMP26), Arithmetic Volumes (FYZ26)	Independence Polynomials (LeeSeo26), Generalized Erdős-1051 (BKKKZ26)	Eigenweights (Feng26)

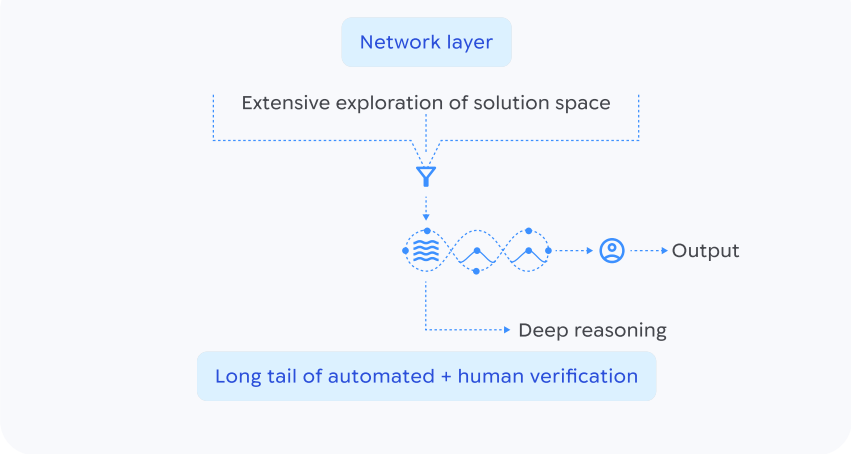
Level 3: Significant Advance			
Level 4: Landmark Breakthrough			

Classification of all AI-assisted mathematics results encompassed in this work. *Works listed as Level 2 in this table have been submitted for publications.

Prompts and model outputs are available here. For discussions on AI contributions, our “Human-AI Interaction card”, and community impact, see our paper.

Expanding to Physics and Computer Science

Gemini Deep Think mode has also demonstrated promise in computer science and physics. The second paper builds on similar agentic reasoning ideas, and identifies effective "recipes" for collaboration, specifically the "Advisor" model, where humans guide AI through iterative "Vibe-Proving" cycles to validate intuition and refine proofs. We also detail tactical techniques like “balanced prompting” —requesting simultaneous proof *or* refutation to prevent confirmation bias —and code-assisted verification. These methods, combined with the model's ability to bridge disparate scientific fields through deep structural connections, are transforming how theoretical research is conducted. This work builds upon our successful deployment of an advanced version of Gemini Deep Think to assist in reviewing CS theory papers for STOC’26 conference.



A schematic overview of the AI reasoning pipeline, illustrating how extensive solution space exploration at the network layer is funneled into structured reasoning and validated by automated and human verification.

Collaborating with experts on 18 research problems, an advanced version of Gemini Deep Think helped resolve long-standing bottlenecks across algorithms, ML and combinatorial optimization, information theory, and economics. Highlights from our "Accelerating Research with Gemini" paper include (corresponding section numbers in paper):

1. **Crossing mathematical borders for network puzzles**: Progress on classic computer science problems like "Max-Cut" (efficiently splitting networks) and the "Steiner Tree" (connecting high-dimensional points) had slowed down. Gemini broke both deadlocks by thinking outside the box. It solved these discrete algorithmic puzzles by pulling advanced tools—like the Kirszbraun Theorem, measure theory, and the Stone-Weierstrass theorem—from entirely unrelated branches of continuous mathematics. See Sections 4.1 and 4.2.

2. **Settling a decade-old conjecture in online submodular optimization**: A 2015 theory paper proposed a seemingly obvious rule for data streams: making a copy of an arriving item is always less valuable than simply moving the original. Experts struggled for a decade to prove this. Gemini engineered a highly specific three-item combinatorial counterexample, rigorously proving the long-standing human intuition false. See Section 3.1.

3. **Machine learning optimization:** Training AI to filter out noise usually requires engineers to manually tune a mathematical "penalty." Researchers created a new technique that did this automatically, but couldn't mathematically explain why. Gemini analyzed the equations and proved the method succeeds by secretly generating its own "adaptive penalty" on the fly. See Section 8.3.

4. **Upgrading economic theory for AI:** A recent 'Revelation Principle' for auctioning AI generation tokens only worked mathematically when bids were restricted to rational numbers. Extending the domain to continuous real numbers invalidated the original proof. Gemini employed advanced topology and order theory to extend the theorem, accommodating real-world, continuous auction dynamics. See Section 8.4.

5. **Physics of cosmic strings:** Calculating gravitational radiation from cosmic strings requires finding analytical solutions to tricky integrals containing "singularities." Gemini found a novel solution using Gegenbauer

polynomials. This naturally absorbed the singularities, collapsing an infinite series into a closed form, finite sum. See Section 6.1.

Spanning diverse fields—from information and complexity theory to cryptography and mechanism design—the results demonstrate how AI is fundamentally shifting research. For details, see our paper.

Given computer science's fluid, conference-driven publication pipeline, we describe these results by academic trajectory rather than a rigid taxonomy. About half target strong conferences —including an ICLR ’26 acceptance—while most remaining findings will form future journal submissions. Even when course-correcting the field by identifying errors (Section 3.2) or refuting conjectures (Section 3.1), these outcomes highlight AI’s value as a high-level scientific collaborator.

The Future of Human-AI Collaboration

Building on Google’s previous breakthroughs (1, 2, 3, 4, 5), this work demonstrates that general foundation models - leveraged with agentic reasoning workflows - can act as a powerful scientific companion.

Under direction from expert mathematicians, physicists, and computer scientists, Gemini Deep Think mode is proving its utility across fields where complex math, logic and reasoning are core.

We are witnessing a fundamental shift in the

scientific workflow. As Gemini evolves, it acts as "force multiplier" for human intellect, handling knowledge retrieval and rigorous verification so scientists can focus on conceptual depth and creative direction. Whether refining proofs, hunting for counterexamples, or linking disconnected fields, AI is becoming a valuable collaborator in the next chapter of scientific progress.

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Authors of the second paper “Accelerating Scientific Research with Gemini: Case Studies and Common Techniques” include David P. Woodruff, Vincent Cohen-Addad, Lalit Jain, Jieming Mao, Song Zuo, MohammadHossein Bateni, Simina Branzei, Michael P. Brenner, Lin Chen, Ying Feng, Lance Fortnow, Gang Fu, Ziyi Guan, Zahra Hadizadeh, Mohammad T. Hajiaghayi, Mahdi JafariRaviz, Adel Javanmard, Karthik C. S., Ken-ichi Kawarabayashi, Ravi Kumar, Silvio Lattanzi, Euiwoong Lee, Yi Li, Ioannis Panageas, Dimitris Paparas, Benjamin Przybocki, Bernardo Subercaseaux, Ola Svensson, Shayan Taherijam, Xuan Wu, Eylon Yogev, Morteza Zadimoghaddam, Samson Zhou, Yossi Matias, Jeff Dean, James Manyika, Vahab Mirrokni. This list includes Google researchers building the agentic reasoning on top of Gemini, and our academic expert collaborators verifying and collaborating with Gemini. We also thank Corinna Cortes for her careful review of the paper.

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